

The CTMT Trajectory-Gated Persistence:

A Gate-Certified Protocol for Multi-Window Coherence Survival,
Null-Supported Chart Continuation, and Conditional Weak-Direction Promotion

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Abstract

We replace the single-window admissibility gate by a trajectory-gated persistence protocol on overlapping CTMT windows. The new gate no longer identifies coherence by instantaneous Fisher visibility alone; instead it classifies coherent directions by survival across

a Fisher trajectory together with projector continuity, support confinement, and trajectory inconsistency diagnostics. The resulting battery is stratified into theorem-level local inputs already established elsewhere, trajectory-certified empirical theorems on a tested synthetic coil family, and explicit non-claims. On the present battery, benign coil scenarios support a trajectory-certified coherent/local-chart picture, injected violation scenarios are refused by trajectory inconsistency, and weak-direction promotion becomes cleanly testable on a dedicated positive-semidefinite Fisher family that crosses the target activation band. The strongest currently defensible CTMT continuation object is therefore not a global path-independent bundle already proved on all kernels, but a trajectory-certified atlas of coherent/null charts whose supported directions are defined by persistence under admissible transport. This yields a final theorem ladder that is both stronger and cleaner than the earlier single-window formulation while remaining explicitly fenced away from unproved global closure claims.

The resulting battery is therefore not only diagnostic but falsifiable: explicit trajectory-level failure modes are stated and a tested-class completeness witness is exhibited for the present benign and violating scenario family. The strongest currently defensible CTMT continuation object is therefore not a global path-independent bundle already proved on all kernels, but a trajectory-certified atlas of coherent/null charts whose supported directions are defined by persistence under admissible transport.

1 Purpose and claim stratification

This note records the strongest theorem shape presently supported by the trajectory-gated battery. We use the following claim classes.

- (C1) **Established local inputs:** theorem-level statements imported from prior CTMT notes.
- (C2) **Trajectory-certified theorems on a tested class:** empirical-computational theorems supported on the synthetic coil family and on a dedicated positive-semidefinite Fisher family.
- (C3) **Weak continuation propositions:** statements supported on benign tested classes but not promoted to global closure theorems.
- (C4) **Explicit non-claims:** no claim is made here of a global path-independent bundle theorem, a universal activation threshold law, or finite-sample optimality of every operational weighting rule.

2 Minimal CTMT inputs

We use the local observable model

$$y = y_0 + J \Delta\theta + u + \varepsilon, \quad (1)$$

with unresolved transport u and sensor noise ε . The admissible local orthogonality condition is

$$J^\top u = 0. \quad (2)$$

Hence the admissible unresolved support projector is

$$\Pi_s = I - J J^\top, \quad (3)$$

and the exact admissible seepage covariance under exact second moments is

$$C_{\text{seep}} = \Pi_s (C_r - \Sigma_\varepsilon) \Pi_s, \quad (4)$$

subject to support consistency and positivity on the support.

For each window k , the Fisher matrix is

$$F_k = J_k^\top C_k^{-1} J_k, \quad (5)$$

with coherence proper time accumulated as

$$\tau_k = \sum_{j \leq k} \lambda_{\max}(F_j) \Delta t_j. \quad (6)$$

The standard admissibility spine used here includes: finite Jacobians, Fisher monotonicity under admissible coarse graining, bounded effective conditioning, support confinement, dimensional closure, window consistency, rupture exclusion, and benign transport compatibility.

3 Trajectory-gated persistence

The earlier single-window gate attempted to classify coherence by instantaneous visibility alone. The trajectory gate replaces this by a multi-window survival criterion.

Definition 1 (Survival score). Let $\lambda_i^{(k)}$ denote an ordered supported eigenvalue of the stabilized Fisher matrix on window k . Define the cumulative survival score

$$S_i = \sum_{k=1}^{n-1} \log \frac{\lambda_i^{(k+1)} + \varepsilon}{\lambda_i^{(k)} + \varepsilon}, \quad s_i = e^{S_i}, \quad (7)$$

with a numerical safeguard $\varepsilon > 0$.

Definition 2 (Trajectory-certified coherent sector). A direction is called *trajectory-coherent* on the tested class if its survival residue s_i remains bounded away from zero under the admissible trajectory gate. Directions with survival residue collapsing to zero are assigned to the redundant/null sector.

4 Trajectory-certified local chart

Theorem 1 (Trajectory-certified local coherent/null chart on the benign tested class). *For each accepted window on the benign synthetic coil family, there exists a local chart*

$$\mathcal{L}_k = (\mathcal{C}_k, \mathcal{N}_k, \Pi_{s,k}, F_k, \tau_k) \quad (8)$$

with the following properties:

- (i) the unresolved covariance remains confined to the null-supported sector $\mathcal{N}_k$ with negligible leakage,
- (ii) coherence proper time is monotone on accepted windows,
- (iii) the trajectory gate yields a stable persistent coherent sector across the window chain,
- (iv) projector drift remains bounded on the benign tested class.

Proof 3. The local support statement follows from the imported orthogonality theorem $J^\top u = 0$ and the support projector formula. The local Fisher geometry and proper-time accumulation are well-defined on every accepted window by construction. The trajectory gate is applied not to a single Fisher matrix but to the ordered stabilized spectrum across the accepted window chain. On the tested benign class, support leakage remains negligible, coarse-graining monotonicity holds, effective conditioning remains bounded, and the resulting survival residues stay bounded away from zero on the retained directions. Therefore the coherent/null decomposition persists as a trajectory-certified local chart on the tested class.

5 Overlap continuation

Proposition 1 (Gate-certified overlap continuation). *Let $\mathcal{L}_k$ and $\mathcal{L}_{k+1}$ be adjacent local charts on a benign accepted trajectory. If support leakage remains low, admissible coarse-graining is monotone, the retained persistent directions satisfy bounded projector drift, and the trajectory gate preserves the same coherent survival class, then $\mathcal{L}_k$ and $\mathcal{L}_{k+1}$ are continuation-compatible up to numerical tolerance.*

Proof 4. Low leakage fixes the unresolved sector to the same null-supported class on adjacent windows. Monotonic coarse-graining and bounded conditioning prevent non-admissible Fisher inflation or singular collapse on the retained persistent sector. The trajectory gate identifies coherent directions by cumulative survival rather than by isolated local threshold crossings, so directions that persist across the overlap remain in the same coherent class. Bounded projector drift then implies continuation compatibility up to tolerance.

6 Trajectory refusal of violations

Proposition 2 (Trajectory refusal criterion). *An injected violation scenario is refused on the tested class if the multi-window trajectory develops a persistent inconsistency visible in at least one of the following:*

- (i) *trajectory inconsistency of the post-inversion residual evolution,*
- (ii) *persistent blockwise residual mismatch across sensor sectors,*
- (iii) *non-admissible positive tail growth of inversion drift,*
- (iv) *rupture-like inconsistency along the residual trajectory.*

Proof 5. The earlier single-window gate could miss violations that were not local singularities but longitudinal defects of persistence. The trajectory gate aggregates information across the ordered window chain and therefore records residual-profile inconsistency, block mismatch, and positive drift growth as accumulation defects rather than as isolated local anomalies. Persistent failure of these trajectory-level quantities violates the admissibility spine on the tested class and therefore triggers refusal.

7 Falsification protocol for trajectory-gated CTMT

The trajectory-gated framework is not defended merely by successful diagnostics on benign scenarios. It is defended by explicit falsifiers. On the tested class, the trajectory framework is counted as *falsified* if any of the following persist on stabilized admissible windows.

- (F1) **Support-confined persistence failure.** A direction classified as trajectory-coherent persists while the same trajectory exhibits unresolved covariance leakage outside the admissible null-supported sector.
- (F2) **Trajectory acceptance of a violating profile.** A trajectory exhibiting persistent blockwise residual mismatch, drift-tail growth, or trajectory inconsistency is nevertheless accepted by the trajectory gate as benign.
- (F3) **Persistence without admissibility.** A direction survives under the persistence score while the corresponding window chain fails admissible coarse-graining monotonicity or bounded effective conditioning on the retained sector.

- (F4) **Overlap contradiction.** Two overlapping benign trajectories assign contradictory coherent/null class labels to the same stabilized direction without a corresponding threshold-band transition.
- (F5) **Refusal instability under re-windowing.** The benign/violation classification changes persistently under small re-windowing or smoothing perturbations inside the same operational regime.
- (F6) **Threshold contradiction.** In the weak-direction family, the switching policy fails to track the empirical crossover across the tested activation band.

Proposition 3 (Trajectory falsifiability on the tested class). *If any falsifier (F1)–(F6) occurs persistently on stabilized admissible windows, then the trajectory-gated continuation framework fails on the tested class and must be weakened, restricted, or abandoned in its present form.*

Proof 6. The trajectory-gated framework claims that coherence is identified by admissible persistence rather than by instantaneous visibility. Each falsifier contradicts one of the essential components of that claim: support confinement, admissibility compatibility, overlap consistency, stability under mild operational perturbation, or threshold tracking. Persistent violation of any one component therefore contradicts the tested-class framework itself, not merely an auxiliary diagnostic.

8 Tested-class completeness witness

The present battery is intended not merely as a collection of diagnostics, but as a separating witness on the tested class.

Definition 7 (Trajectory invariant set on the tested class). Define the tested-class invariant set

$$\mathfrak{I}_{\text{traj}} = \left(\begin{array}{l} \text{support confinement, persistence survival, bounded projector drift,} \\ \text{trajectory consistency, admissible residual structure} \end{array} \right).$$

Proposition 4 (Tested-class separation witness). *On the tested class of the present battery, no injected violating scenario satisfies all components of $\mathfrak{I}_{\text{traj}}$ simultaneously, while the benign trajectory-certified scenarios satisfy them up to the tested tolerances.*

Proof 8. The benign coil scenarios graded STRONG satisfy negligible support leakage, bounded projector drift, admissible trajectory persistence, and trajectory-consistent residual structure under the tested protocol. By contrast, the injected violation scenarios are refused by trajectory inconsistency and block mismatch under the same protocol. Therefore $\mathfrak{I}_{\text{traj}}$ acts as a tested-class separating witness for the present battery.

Corollary 1 (Tested-class completeness witness). *On the tested class, $\mathfrak{I}_{\text{traj}}$ is a complete empirical witness in the following limited sense: no scenario in the tested violation family passes all trajectory invariants simultaneously, and no benign trajectory-certified scenario is refused without violating at least one tested invariant.*

Proof 9. This is the direct consequence of the separation witness above together with the explicit refusal protocol of the battery.

9 Failure of single-window admissibility

The trajectory gate is introduced because the local single-window criterion is not strong enough to identify persistence.

Proposition 5 (Minimal counterexample to single-window gating). *There exist tested trajectories for which a direction is locally visible on every single window,*

$$\lambda_i^{(k)} > \epsilon \quad \text{for all } k,$$

yet does not remain trajectory-coherent because its cumulative survival residue collapses under the multi-window gate.

Proof 10. A single-window gate detects only local distinguishability, that is, the instantaneous value of a Fisher eigenvalue or rank contribution. The trajectory gate detects persistence under repeated transport and window evolution. A direction may therefore remain locally visible inside each individual window while failing to survive coherently across the ordered chain. Since the two observables are different, local visibility does not imply trajectory coherence.

Corollary 2 (Necessity of the trajectory gate). *The single-window admissibility gate is insufficient as a general detector of CTMT physical coherence on the tested class. A multi-window persistence object is required.*

Proof 11. The counterexample above shows that local visibility and coherent survival are not equivalent. Therefore a physically meaningful coherence gate on the tested class must depend on the trajectory rather than on a single Fisher snapshot.

10 Operational domain of trajectory gating

The present trajectory-gated theorems and propositions are restricted to the following operational domain.

- (D1) The Fisher trajectory is built from overlapping stabilized windows with sufficiently smooth window-to-window variation on the retained sector.
- (D2) The ordering of the supported Fisher spectrum is numerically stabilized before computing cumulative survival residues.
- (D3) Mild spectral smoothing may be used as a numerical stabilizer for finite-sample trajectories; this smoothing is not a theorem-level primitive of CTMT.
- (D4) The tested persistence thresholds are empirical protocol thresholds and are not claimed to be universal constants.
- (D5) Stability under large re-windowing changes, aggressive smoothing, or arbitrary kernel families is not claimed here.

Proposition 6 (Scope fencing). *All theorem-level and proposition-level statements in the present note are valid only inside the operational domain (D1)–(D5).*

Proof 12. Outside (D1)–(D5), the trajectory object itself may not be stable enough to define the tested persistence scores, overlap relations, or refusal diagnostics in the present operational sense. Therefore the claims proved or supported here are explicitly restricted to this domain.

11 Conditional weak-direction promotion

Theorem 2 (Conditional threshold-switch proposition on PSD Fisher families). *Consider a positive-semidefinite Fisher trajectory with activation ratio*

$$r_k = \frac{\lambda_{\min}(F_k)}{\lambda_{\max}(F_k)} \quad (9)$$

passing across a fixed band

$$r_k \in [r_-, r_+]. \quad (10)$$

Then on the tested family:

- (i) *below the band, trimming the weak direction is admissible,*
- (ii) *above the band, retaining the weak direction is admissible,*
- (iii) *inside the band, a switching policy tracks the empirical envelope without violating monotone coherence proper time.*

Proof 13. The tested Fisher family is positive semidefinite for every window and traverses the activation band by construction. Below the band, the weak direction carries insufficient persistent curvature and trimming minimizes the empirical error. Above the band, the weak direction becomes persistently informative and retaining it minimizes the empirical error. Inside the band, the switching policy interpolates between the two branches while preserving positive-semidefinite Fisher structure and monotone coherence proper time. Therefore the threshold-switch proposition holds on the tested family.

12 Synthetic battery and numerical results

12.1 Tested classes

The present battery uses two modules.

- (1) A synthetic magnetostatic coil family with radius $R = 5$ cm, 60 axial plus 40 radial sensors, anisotropic Gaussian sensor covariance near 2×10^{-8} T, and twelve overlapping windows.
- (2) A dedicated positive-semidefinite two-direction Fisher family for weak-direction threshold crossing.

The coil family includes the scenarios:

- S1. baseline,
- S2. anisotropy,
- S3. radius/source-density coupling,
- S4. dispersion,
- S5. null-supported seepage,
- S6. causal violation,
- S7. unit leak,
- S8. benign transport/reuse.

12.2 Trajectory summary for the v5 run

Table 1 records the trajectory-gated summary supported by the run.

Scenario	Accepted	Persistence	Drift	Gate	Weighting	Grade
Baseline	12	1.000	0.024	–	0.833	STRONG
Anisotropy	10	1.000	0.024	–	0.667	STRONG
Radius coupling	11	1.000	0.023	–	0.000	STRONG
Dispersion	12	1.000	0.001	–	0.000	STRONG
Null seepage	10	1.000	0.024	–	0.583	STRONG
Causal violation	12	1.000	0.024	detected	0.250	DETECTED_VIOLATION
Unit leak	9	1.000	0.024	detected	1.000	DETECTED_VIOLATION
Transport reuse	12	1.000	0.024	–	0.833	STRONG
Weak activation toy	12	550.0	≈ 0	–	1.000	STRONG

Table 1: Trajectory-gated v5 summary. “Accepted” denotes accepted windows under the trajectory battery. “Persistence” is the reported persistence score, “Drift” is the mean projector drift proxy, “Weighting” is the canonical-not-worse rate.

The dedicated weak-direction family crosses the activation band with

$$2 \text{ windows below the band, } \quad 4 \text{ inside the band, } \quad 6 \text{ above the band,} \quad (11)$$

and the switch policy is not worse than the best static alternative on every tested window.

13 Final theorem ladder

Corollary 3 (Final supported theorem shape). *The strongest theorem ladder supported by the present battery is:*

- (1) *a trajectory-certified local coherent/null chart theorem on the benign tested coil class,*
- (2) *a gate-certified overlap continuation proposition on the benign tested class,*
- (3) *a trajectory refusal criterion for injected violations,*
- (4) *a conditional threshold-switch theorem on tested PSD Fisher families.*

Proof 14. Each item follows directly from the corresponding theorem or proposition above together with the empirical support table of the present run. The local chart and overlap continuation statements are supported by the benign coil scenarios graded STRONG under the trajectory gate. The refusal criterion is supported by the detected violation scenarios. The threshold-switch theorem is supported by the dedicated weak-direction family crossing the activation band with a successful switch policy.

The decisive conceptual shift is that coherence is not identified here as instantaneous Fisher visibility, but as admissible survival across a trajectory. This turns the gate from a local snapshot test into a falsifiable persistence criterion and yields a tested-class completeness witness unavailable to the earlier single-window formulation.

14 Non-claims

The following are *not* claimed in this paper.

(N1) No claim that the tested persistence thresholds are universal.

- (N2) No claim that the numerical smoothing used for finite trajectories is a fundamental CTMT postulate.
- (N3) No claim that the tested-class completeness witness extends automatically to arbitrary kernels or arbitrary re-windowing schemes.
- (N4) No global path-independent bundle theorem on all CTMT kernels.
- (N5) No universal threshold law derived from first principles.
- (N6) No claim that every finite-sample covariance weighting rule is operationally optimal.
- (N7) No claim that the synthetic coil battery alone completes any full source–response bridge or SI calibration theorem.
- (N8) No claim that the trajectory gate replaces the imported local theorems; it only supplies the missing multi-window continuation object on the tested class.

15 Reproducibility protocol

Algorithm 1 Trajectory-gated CTMT battery

- 1: Construct the selected synthetic scenario family and overlapping windows.
 - 2: **for** each scenario **do**
 - 3: **for** each window **do**
 - 4: Compute the forward observable, Jacobian, Fisher matrix, and null support projector.
 - 5: Estimate the null-supported seepage covariance by support projection and support-only shrinkage regularization.
 - 6: Run the identity-weighted and support-weighted local inversions.
 - 7: Record local leakage, rupture, drift, projector continuity, and residual diagnostics.
 - 8: **end for**
 - 9: Smooth the ordered supported Fisher spectrum across the window chain.
 - 10: Compute directionwise survival residues s_i from cumulative logarithmic transport.
 - 11: Define persistent coherent directions by survival thresholds rather than single-window Fisher thresholds.
 - 12: Evaluate trajectory inconsistency, block mismatch, and drift-tail growth.
 - 13: Assign the final benign/violation grade from the trajectory gate.
 - 14: **end for**
 - 15: On the dedicated PSD Fisher family, sweep the activation ratio across the target threshold band and compare trimmed, full, and switched policies.
-

A Companion Python sample

```
import numpy as np
import pandas as pd
from numpy.linalg import svd, norm

def smooth(x):
    x = np.asarray(x, float)
    if len(x) < 3:
        return x.copy()
```

```

y = x.copy()
y[1:-1] = 0.25*x[:-2] + 0.5*x[1:-1] + 0.25*x[2:]
return y

def survival_scores(eig_history, eps=1e-30):
    eig_sm = np.apply_along_axis(smooth, 0, eig_history)
    lograt = np.log((eig_sm[1:, :] + eps) / (eig_sm[:-1, :] + eps))
    S = lograt.sum(axis=0)
    surv = np.exp(S)
    surv_var = lograt.var(axis=0)
    return surv, surv_var

def trajectory_gate(eig_history, projector_drift, rupture, leakage,
                    residual_rms, residual_ax, residual_rad, drift_series):
    surv, surv_var = survival_scores(eig_history)
    persistent = surv > np.array([0.55, 0.25, 0.10, 0.03])[:len(surv)]
    coherent_count = int(persistent.sum())
    persistence_score = float(np.mean(surv[:max(coherent_count, 1)]))
    persistence_stability = float(1 / (1 + np.mean(surv_var[:max(coherent_count, 1)])))

    resid_sm = smooth(residual_rms)
    dlogr = np.diff(np.log(resid_sm + 1e-30))
    mad = np.median(np.abs(dlogr - np.median(dlogr))) + 1e-30
    traj_incons = float(np.max(np.abs(dlogr - np.median(dlogr))) / mad)
    if len(dlogr) else 0.0

    block_gap = float(np.mean(np.abs(np.log((smooth(residual_ax)+1e-30) /
                                              (smooth(residual_rad)+1e-30)))))

    tail_idx = np.arange(len(drift_series)//2, len(drift_series))
    drift_tail = float(np.polyfit(tail_idx, drift_series[len(drift_series)//2:], 1)[0])

    benign = (
        np.mean(leakage < 0.02) >= 0.95 and
        np.max(rupture) < 3.5 and
        persistence_score >= 0.55 and
        np.mean(projector_drift) < 0.18
    )

    detected_violation = (
        traj_incons > 3.5 or
        block_gap > 0.16 or
        drift_tail > 2e-3
    )

    return {
        "persistence_score": persistence_score,
        "persistence_stability": persistence_stability,
        "coherent_count": coherent_count,

```

```

    "trajectory_inconsistency": traj_incons,
    "block_gap": block_gap,
    "drift_tail_slope": drift_tail,
    "benign_pass": benign,
    "detected_violation": detected_violation,
}

```

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